

强关联费米子的非平衡临界动力学

余荫铠 (Yin-Kai Yu)

✉ yuyinkai@iphy.ac.cn | 🏠 www.yykspace.com | 🏛️ 中国科学院物理研究所

YKY, Zhi Zeng, Yu-Rong Shu, Zi-Xiang Li, Shuai Yin, arXiv:2310.10601 (2023)

YKY, Zhi-Xuan Li, Shuai Yin, Zi-Xiang Li, arXiv:2410.18854 (2024)

Zhi Zeng, **YKY**, Zhi-Xuan Li, Zi-Xiang Li, Shuai Yin, Nature Communications 16, 6181 (2025).

Zhi Zeng, **YKY**, Zi-Xiang Li, Shuai Yin, Physical Review B 112, L060301 (2025).

2025 年 10 月 15 日 广州

领域の动机

- **费米子**：无经典对应的费米子统计/泡利不相容
- **强关联**：单体近似（能带论、费米液体、DFT、对角化）失效，真·多体理论
- **量子临界性**：标度性与普适性，发散的涨落模式（无穷长时间演化才能平衡）

「平衡态只是特例，非平衡才是常态」

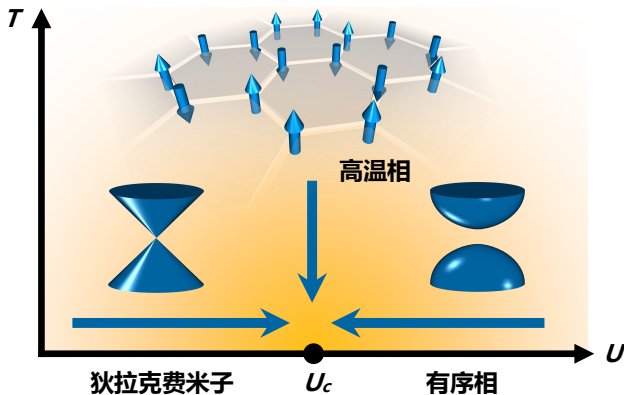
如何建立强关联费米子的非平衡临界动力学理论？



系统性的开篇

Dirac 量子临界性中的非平衡动力学¹

$$H = -t \sum_{\langle ij \rangle, \sigma} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i \left(n_{i\uparrow} - \frac{1}{2} \right) \left(n_{i\downarrow} - \frac{1}{2} \right)$$



¹Yin-Kai Yu, Zhi Zeng, Yu-Rong Shu, Zi-Xiang Li, and Shuai Yin, arXiv:2310.10601, (2023).

系统性的开篇

重整化群视角

重整化群变换 $a \rightarrow ba, \Lambda \rightarrow \Lambda/b$, 标度假定

$$\langle O(U) \rangle = b^{-\kappa} f\left((U - U_c)b^{1/\nu}\right)$$

有限尺寸标度²

$$\langle O(U, L) \rangle = b^{-\kappa} f\left((U - U_c)b^{1/\nu}, Lb^{-1}\right)$$

非平衡标度³⁴⁵

$$\langle O(U, L, \tau) \rangle = b^{-\kappa} f\left((U - U_c)b^{1/\nu}, Lb^{-1}, \tau b^{-z}\right)$$

取 $b = \tau^{1/z}$

$$\langle O(U, L, \tau) \rangle = \tau^{-\kappa/z} f\left((U - U_c)\tau^{1/\nu z}, L\tau^{-1/z}\right)$$

²Fisher, Reports on progress in physics, (1967).

³Janssen, Schaub, and Schmittmann, Z. Phys. B, (1989).

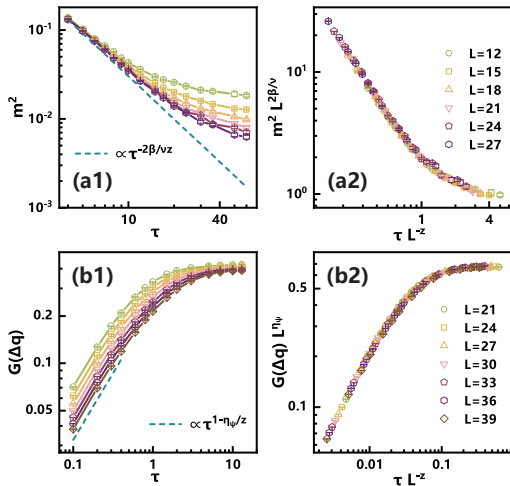
⁴Yin, Mai, and Zhong, PRL, (2014).

⁵Yin-Kai Yu, Zhi Zeng, Yu-Rong Shu, Zi-Xiang Li, and Shuai Yin, arXiv:2310.10601, (2023).



系统性的开篇

非平衡量子蒙特卡洛数值结果⁶



$$m^2(\tau, L) = \tau^{-2\beta/\nu z} f_m(\tau L^{-z})$$

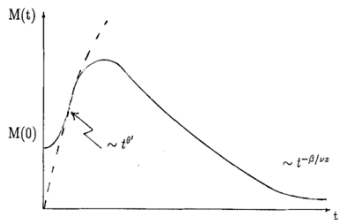
$$G(\tau, L) = \tau^{-\eta_\psi/z} f_G(\tau L^{-z})$$

- 理论框架：从非平衡动力学提取强关联费米子系统的临界点、临界指数
- 两种涨落具有不同的时间尺度

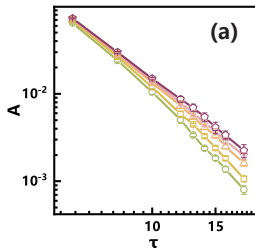
⁶Yin-Kai Yu, Zhi Zeng, Yu-Rong Shu, Zi-Xiang Li, and Shuai Yin, arXiv:2310.10601, (2023).

系统性的开篇

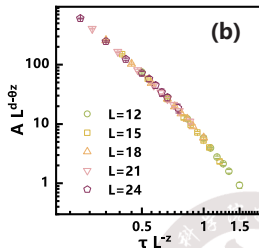
非平衡量子蒙特卡洛数值结果⁷



经典系统/玻色系统: $\theta > 0$



费米系统: $\theta < 0$

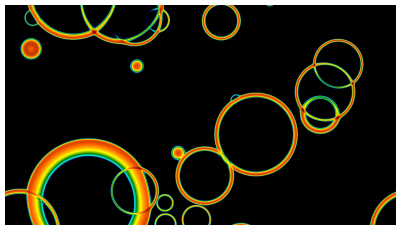


- 临界初始滑移指数 θ , 费米子涨落导致的 $\theta < 0$

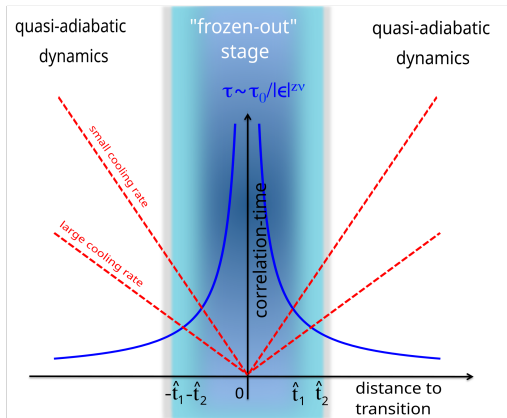
⁷Yin-Kai Yu, Zhi Zeng, Yu-Rong Shu, Zi-Xiang Li, and Shuai Yin, arXiv:2310.10601, (2023).

费米子宇宙中的泡泡

Kibble-Zurek 机制⁸⁹



宇宙相变的拓扑缺陷



“绝热-冻结”范式
关联长度/关联时间的标度

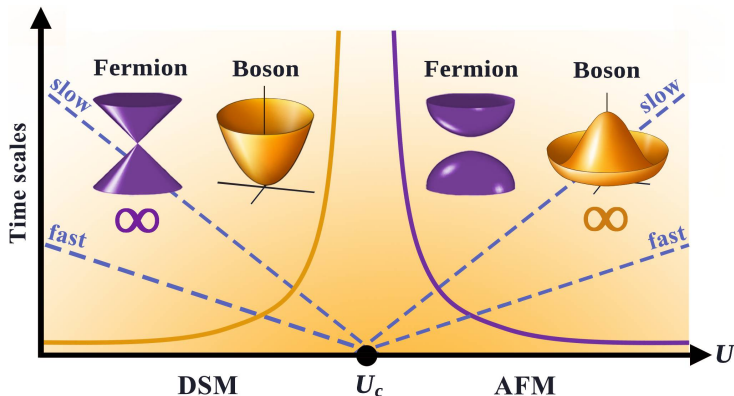
⁸Kibble, Journal of Physics A: Mathematical and General, (1976).

⁹Zurek, Nature, (1985).

费米子宇宙中的泡泡

强关联 Dirac 系统的驱动临界动力学¹⁰

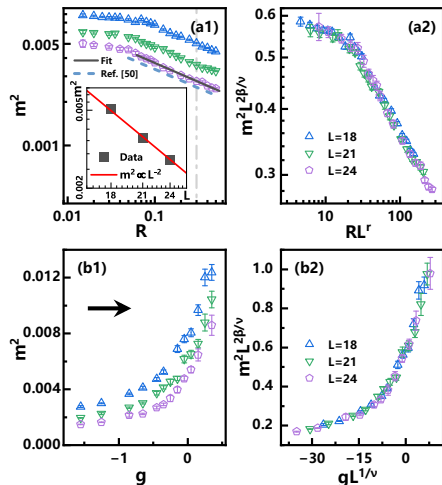
$$H = -t \sum_{\langle ij \rangle, \sigma} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i \left(n_{i\uparrow} - \frac{1}{2} \right) \left(n_{i\downarrow} - \frac{1}{2} \right)$$



¹⁰Zeng, Yu, Li, Li, and Yin, Nature Communications, (2025).

费米子宇宙中的泡泡

超越 Kibble-Zurek 前提的有限时间标度¹¹



$$m^2 = L^{-d} R^{(2\beta-d\nu)/\nu r} \mathcal{F}(RL^r, gL^{1/\nu})$$

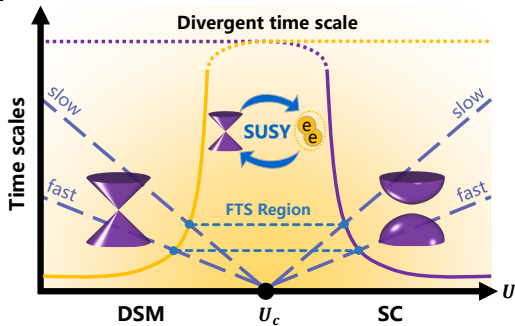
● 更完整的有限时间标度

● 广义判据: $z' < r, r = z + \frac{1}{\nu}$

¹¹Zeng, Yu, Li, Li, and Yin, Nature Communications, (2025).

非平衡动力学也超对称了

演生超对称性¹²

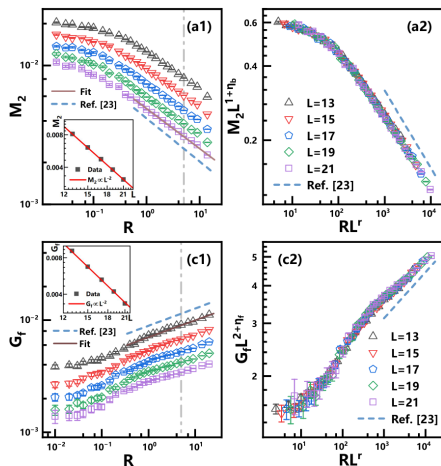


- 单狄拉克锥 Hubbard: $H = \sum_p c_p^\dagger \not{p} c_p + U \sum_i (n_{i\uparrow} - \frac{1}{2})(n_{i\downarrow} - \frac{1}{2})$
- 低能有效理论: $S = \int d^3x \left[\bar{\psi} \not{\partial} \psi + g \left(\phi \psi^T \sigma^y \psi + H.c. \right) + |\partial_\tau \phi|^2 + v_b^2 |\vec{\nabla} \phi|^2 + r |\phi|^2 + u |\phi|^4 \right]$
- 超对称性: $\delta \phi = \bar{\epsilon} \psi, \quad \delta \psi = [-\not{\partial} \phi + g \phi^2] \epsilon$

¹²Li, Vaezi, Mendl, and Yao, Science Advances, (2018).

非平衡动力学也超对称了

具有演生超对称性的非平衡临界动力学¹³

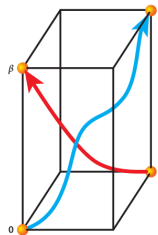


- $M_2 \propto L^{-d} R^{\frac{1+\eta_b-d}{r}}$
- $G_f \propto L^{-d-z+1} R^{\frac{\eta_f}{r}}$
- 超对称性 $\eta_b = \eta_f = \frac{1}{3}$
- 超对称性的实验观测方案

¹³Zeng, Yu, Li, and Yin, Phys. Rev. B, (2025).

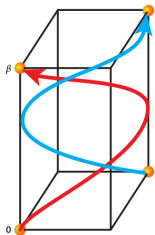
费米统计沉睡时

费米统计



奇数次交换费米子

sign = -1



偶数次交换费米子

sign = +1

$$Z = \text{tr} \left(e^{-\beta H} \right) = \sum_c \det B_c$$

d 维量子系统 = $d + 1$ 维经典系统 ?

符号问题: QMC 采样概率 $\det B_c$ 非正定

强行采样: $\det B_c = \text{sign}_c |\det B_c| \Rightarrow$ 指数级时间复杂度¹⁴, NP-hard

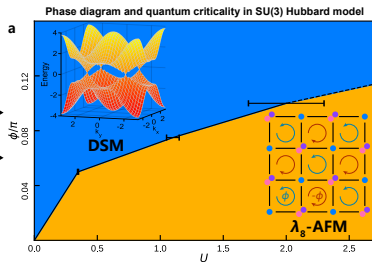
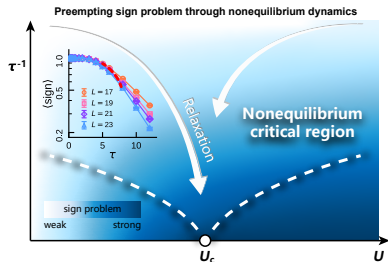
$$\Delta \langle O \rangle \sim \frac{1}{\langle \text{sign}_c \rangle} \sim e^{\alpha \tau}$$

¹⁴Troyer and Wiese, PRL, (2005).



费米统计沉睡时

抢断费米子符号问题¹⁵



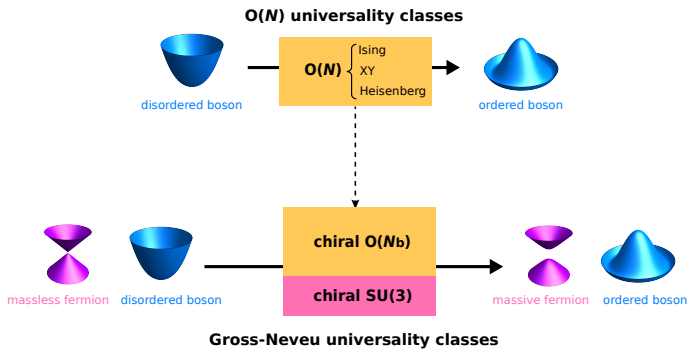
- 趁着符号问题还没反应过来“抢先”把基态相图/量子临界性搞清楚
- 首次求解 SU(3) Hubbard 模型, **"Faster is different"**
- 发现新物相: λ_8 -反铁磁相; 发现新普适类: chiral SU(3)

¹⁵Yin-Kai Yu, Zhi-Xuan Li, Shuai Yin, and Zi-Xiang Li, arXiv:2410.18854, (2024).

费米统计沉睡时

抢断费米子符号问题¹⁶

- 费米子相变普适理论的新理解——



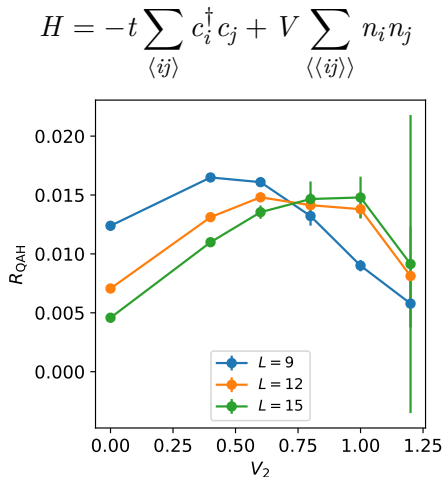
$$\mathcal{L} = \text{tr}(\bar{\psi}\not{\partial}\psi) + g \text{tr}(\bar{\psi}\Phi\psi) + \text{tr}\left[(\partial_\mu\Phi)^2\right] + r \text{tr}(\Phi^2) + \lambda \left[\text{tr}(\Phi^2)\right]^2 + a \left[\text{tr}(\Phi^3)\right]^2 + c \left[\text{tr}(\Phi^2)\right]^4,$$

¹⁶Yin-Kai Yu, Zhi-Xuan Li, Shuai Yin, and Zi-Xiang Li, arXiv:2410.18854, (2024).

费米统计沉睡时

新的进展

- 强关联驱动的量子反常霍尔效应……



总结

费米子为非平衡领域带来了什么？

- Dirac 量子临界性的非平衡动力学框架（弛豫¹⁷、驱动¹⁸）
- 独特的非平衡时间尺度与序参量初始滑移¹⁷
- 超越 Kibble-Zurek 前提的有限时间标度¹⁸
- 非平衡动力学也超对称了¹⁹
-

非平衡方法为费米子研究带来了什么？

- 费米统计沉睡时——抢断费米子符号问题²⁰
- 求解 SU(3) Hubbard 模型基态相图²⁰
- 费米子相变普适理论的新理解²⁰
-

¹⁷Yin-Kai Yu, Zhi Zeng, Yu-Rong Shu, Zi-Xiang Li, and Shuai Yin, arXiv:2310.10601, (2023).

¹⁸Zeng, Yu, Li, Li, and Yin, Nature Communications, (2025).

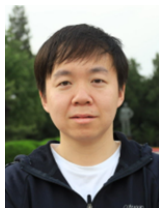
¹⁹Zeng, Yu, Li, and Yin, Phys. Rev. B, (2025).

²⁰Yin-Kai Yu, Zhi-Xuan Li, Shuai Yin, and Zi-Xiang Li, arXiv:2410.18854, (2024).



研究团队

中科院物理所：李自翔、余荫铠



中山大学：阴帅、黎智轩



Imperial College London：曾植



广州大学：舒玉蓉



THANKS!

强关联费米子的非平衡临界动力学

余荫铠 (Yin-Kai Yu)

✉ yuyinkai@iphy.ac.cn | 🏠 www.yykspace.com | 🏢 中国科学院物理研究所

YKY, Zhi Zeng, Yu-Rong Shu, Zi-Xiang Li, Shuai Yin, arXiv:2310.10601 (2023)

YKY, Zhi-Xuan Li, Shuai Yin, Zi-Xiang Li, arXiv:2410.18854 (2024)

Zhi Zeng, **YKY**, Zhi-Xuan Li, Zi-Xiang Li, Shuai Yin, Nature Communications 16, 6181 (2025).

Zhi Zeng, **YKY**, Zi-Xiang Li, Shuai Yin, Physical Review B 112, L060301 (2025).

2025 年 10 月 15 日 广州